
Fraudulent Claim Detection Case Study

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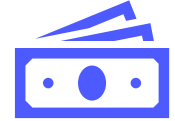
Problem Statement



High volume of claims includes a **notable number of frauds**



Manual detection process is slow and resource-intensive



Fraud often identified **after payouts are made**



Leads to **significant financial losses** for the company



Need for an **early, accurate fraud detection system**



Goal: Use **data-driven models** to classify claims proactively

Business Objective/Goal

01

Build a **predictive model** to classify insurance claims as *fraudulent* or *legitimate*.

02

Use **historical claim data** and customer profiles for training.

03

Leverage features like **claim amount, incident type, and customer demographics**.

04

Detect **fraud patterns** early in the approval process.

05

Answer key questions: What drives fraud? Can we predict it reliably? .

06

Provide **actionable insights** to enhance fraud detection and reduce losses.

Assumptions

Deleted columns
with null values.

Replaced null values
with the
median/mode.

Made some new
features based on
business
understanding.

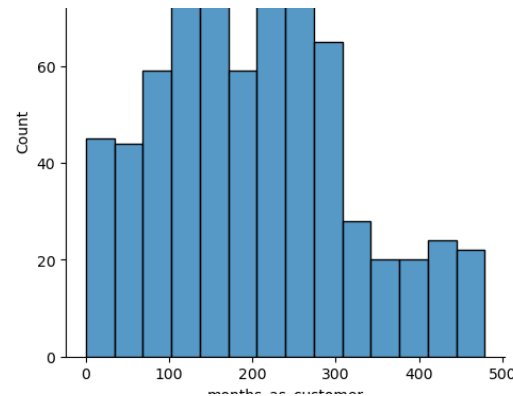
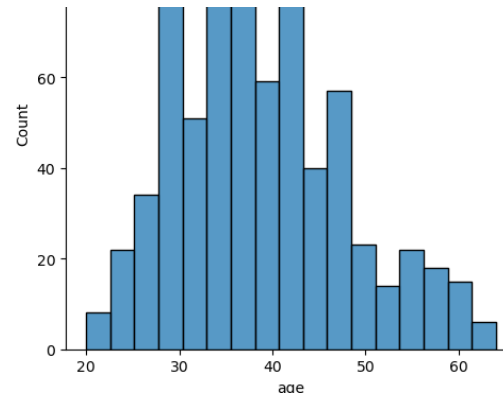
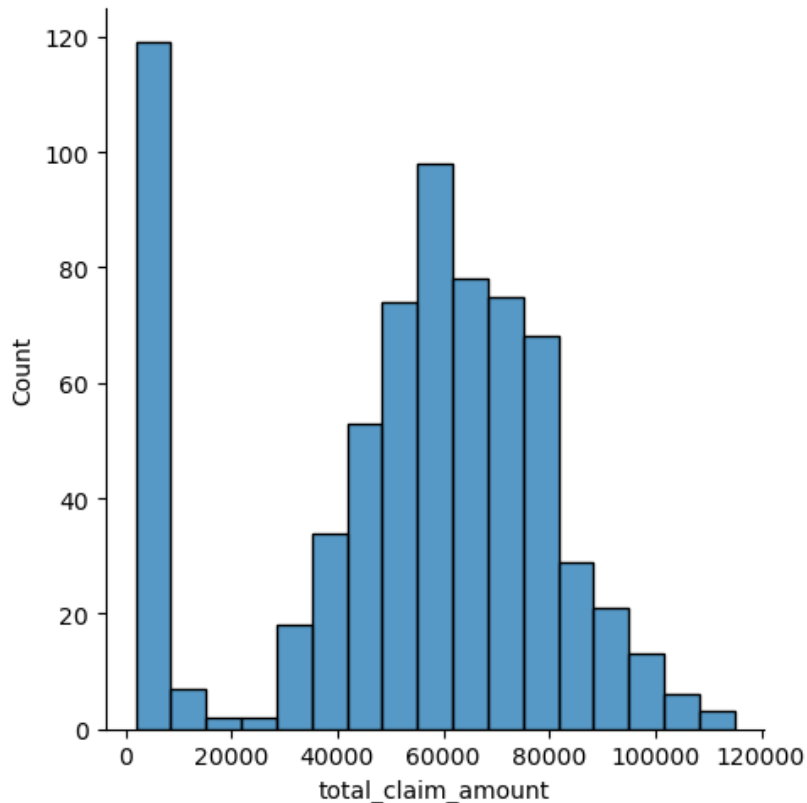
Handled the
incorrect or missing
values.

Dataset overview



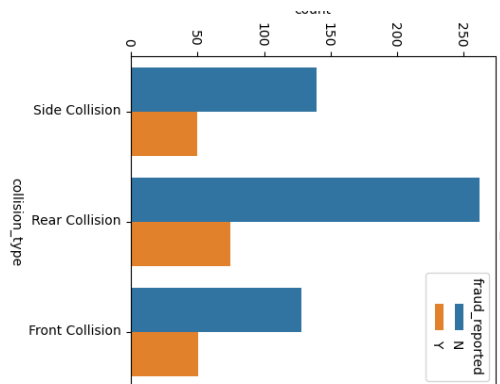
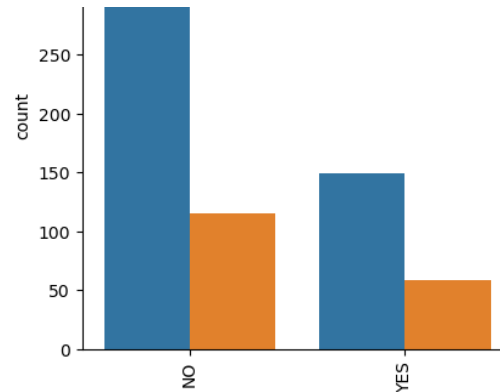
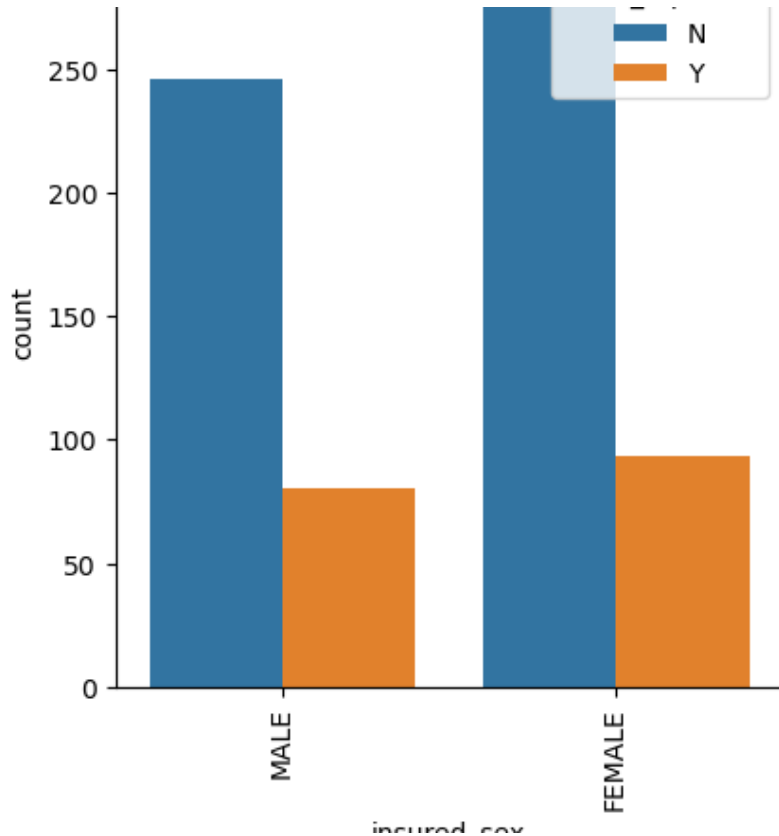
-
- **1000 records** and **40 features** from real-world insurance claims
 - Covers **customer demographics**, **policy details**, **claim info**, and **vehicle specs**
 - Target variable: **fraud_reported (Y/N)**
 - Handled **missing values** and cleaned inconsistent entries
 - Removed **redundant/identifier columns** and fixed data types
 - Addressed **class imbalance** and performed **feature extraction** for model readiness

Exploratory Data Analysis (EDA)- Univariate



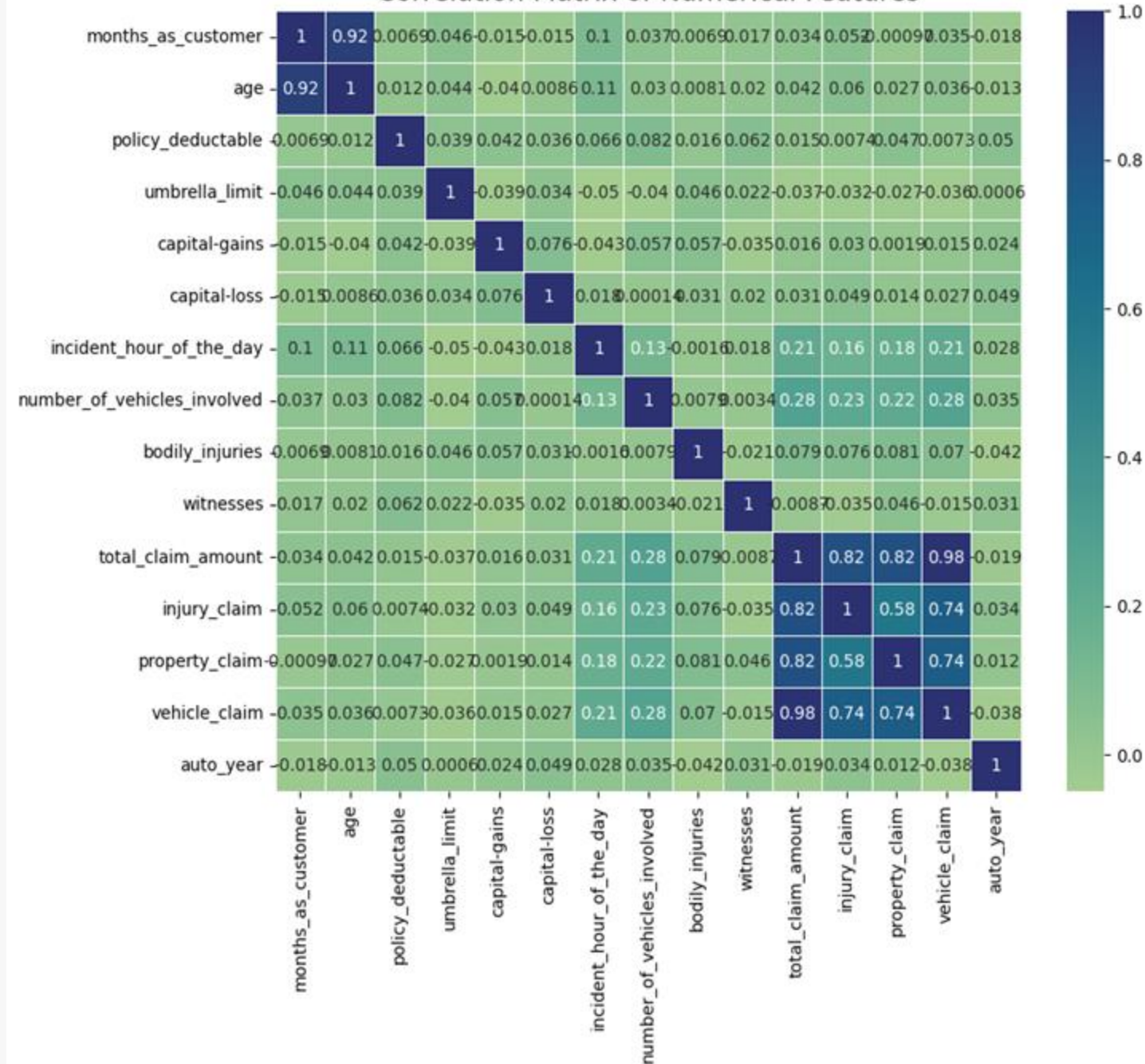
- **Univariate EDA helps** reveal variable distributions, skewness, and potential feature transformations
- **Months as Customer:** Right-skewed; majority between 100–300 months, with few long-tenured outliers
- **Age:** Most policyholders are between 30 and **30–45 years**, forming a mid-age customer base
- **Capital Gains:** Highly skewed; most have **no gains**; a few show **large outliers**
- **Total Claim Amount:** Near-normal, peaked at **lower claims (~\$0)**; most between **\$30K–\$80K**
- **Outlier Detection:** Identified extreme values to improve scaling and modeling

Exploratory Data Analysis (EDA)- Bivariate



- **Bivariate analysis** explores relationships between two variables to uncover meaningful patterns and potential predictors of fraud.
- **insured_sex vs fraud_reported**: Slightly more fraud reported by **females**; gender alone is **not a strong predictor but is** still relevant.
- **collision_type vs fraud_reported**: **Rear collisions** had the most claims, including fraud; may aid fraud detection when **combined with other variables**.
- **property_damage vs fraud_reported**: Most fraud is reported when **property damage = 'NO'**; may indicate **inconsistencies or manipulation**.
- **Detecting fraud** often requires combining multiple features, as individual variables may not show clear patterns alone.
- Bivariate EDA supports **feature selection** and reveals **interactions** that improve model performance.

Correlation Matrix of Numerical Features



Exploratory Data Analysis (EDA)- Multivariate

- Heatmap visualization helped identify **redundant variables** and potential **feature reduction opportunities**.
- Age and months_as_customer** show a strong positive correlation (0.92), suggesting older customers are long-term policyholders.
- A **correlation heatmap** revealed strong multicollinearity among total_claim_amount, vehicle_claim, injury_claim, and property_claim.
- Features like umbrella_limit and capital-gains show minimal correlation, indicating low predictive value.
- High correlations** (e.g., total_claim_amount with vehicle, injury, and property claims) highlight potential **redundant features**.

Methodology

- **Data preparation:**

- Separating the dependent and independent variables.
- Train and validation split:
 1. Splitting the data between the train and validation datasets in a 70:30 ratio.
 2. Resetting the index of new data frames.
- Resampling the training data set.
- Creating new features based on existing features.
- Creating dummies for categorical variables using one-hot encoding.

Logistic regression model:

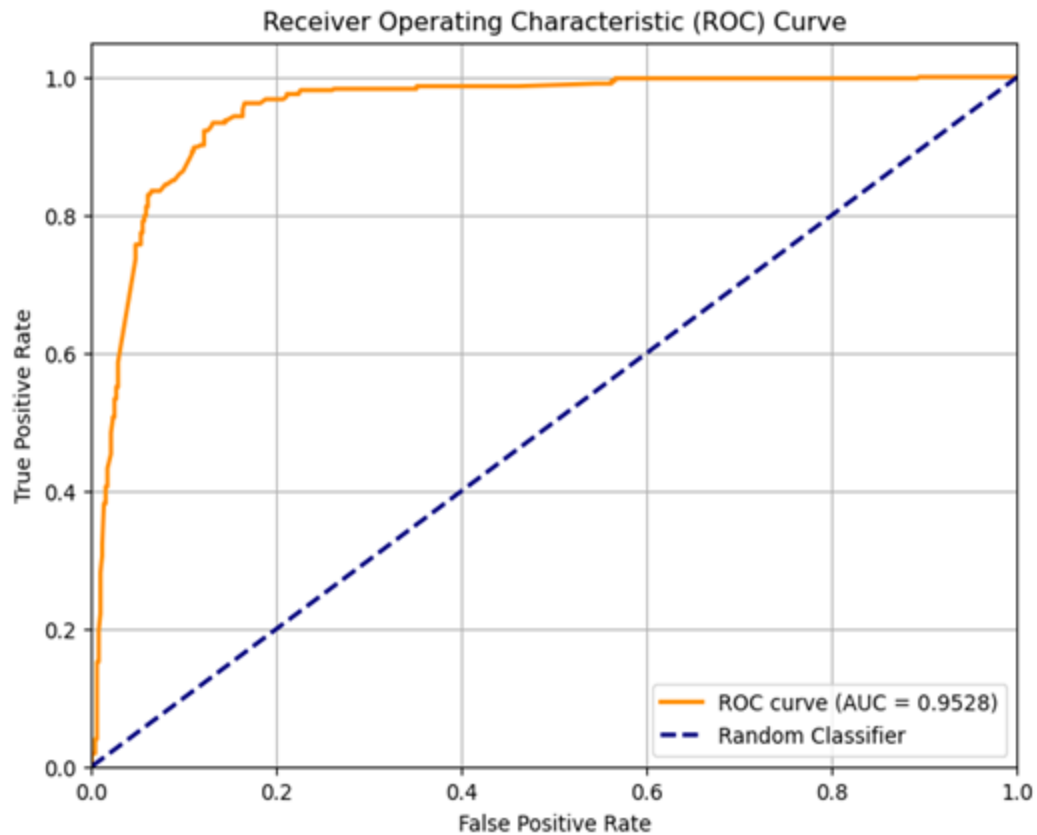
- Selecting the important feature using Recursive Feature Elimination with Cross-Validation (RFECV).
- Manually removing features masked on p-values and VIF values.
- Find the optimal cutoff using scores of the model, then test on validation data.

Random Forest:

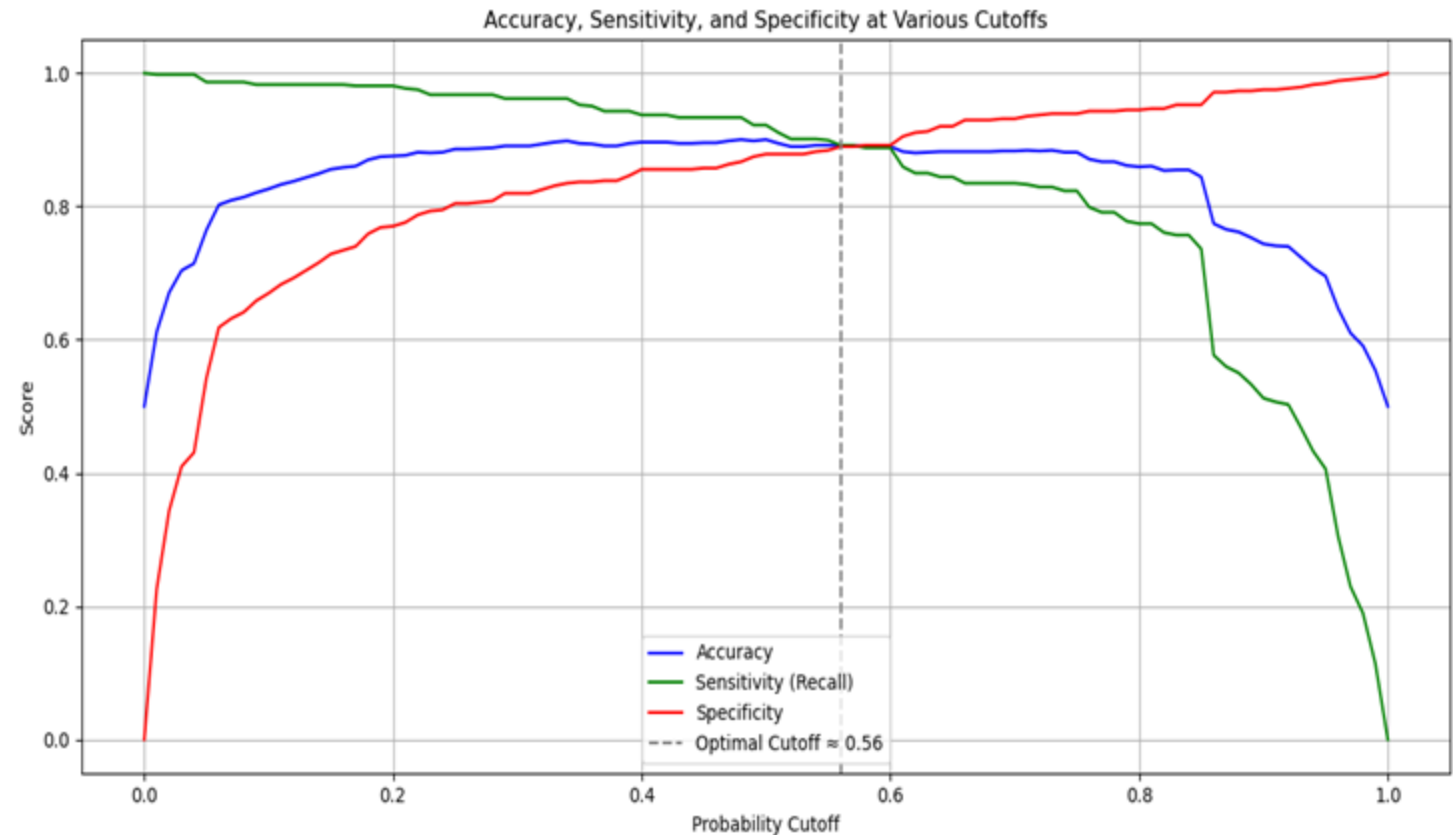
- Make a base model on important features.
- Hypertune to find the best fit, then make the best-fit model.
- Use this model on validation data set and get the scores like recall, F1, accuracy

Methodology

ROC-AUC curve

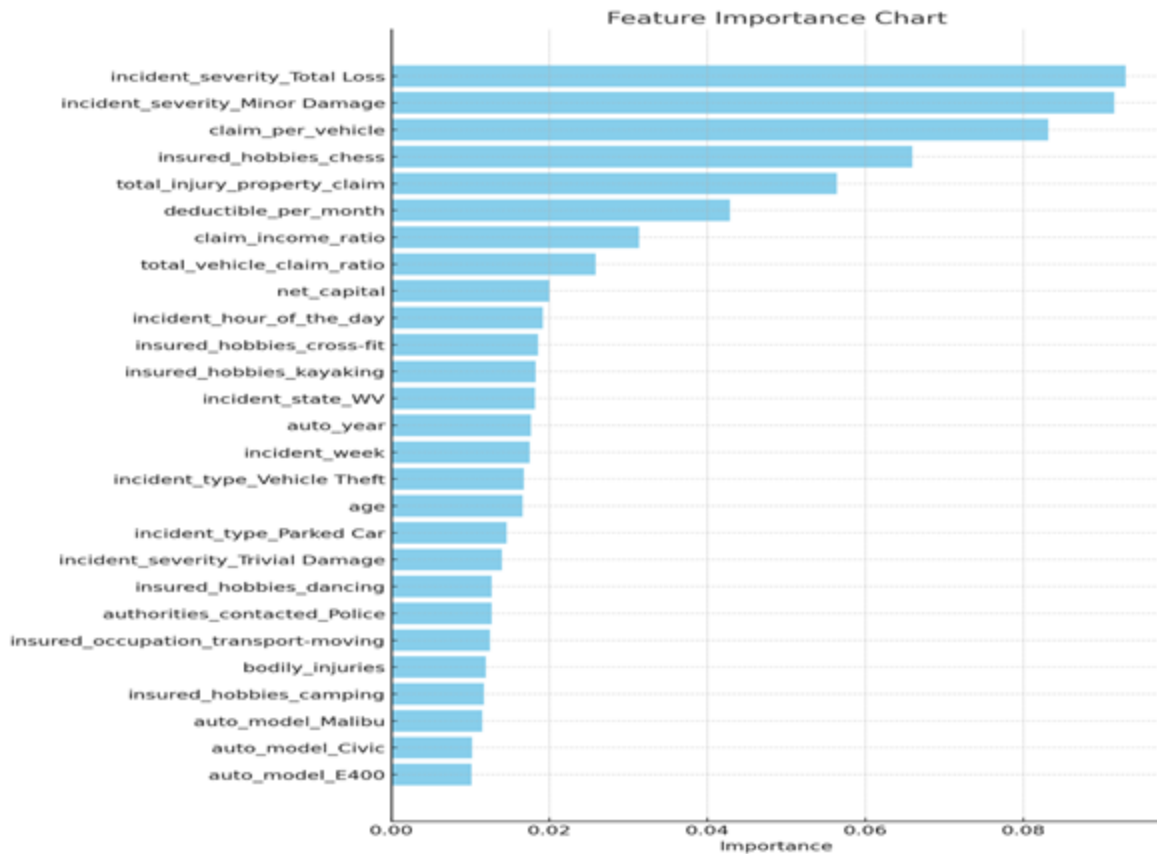


Optimal cutoff using accuracy scores



Methodology

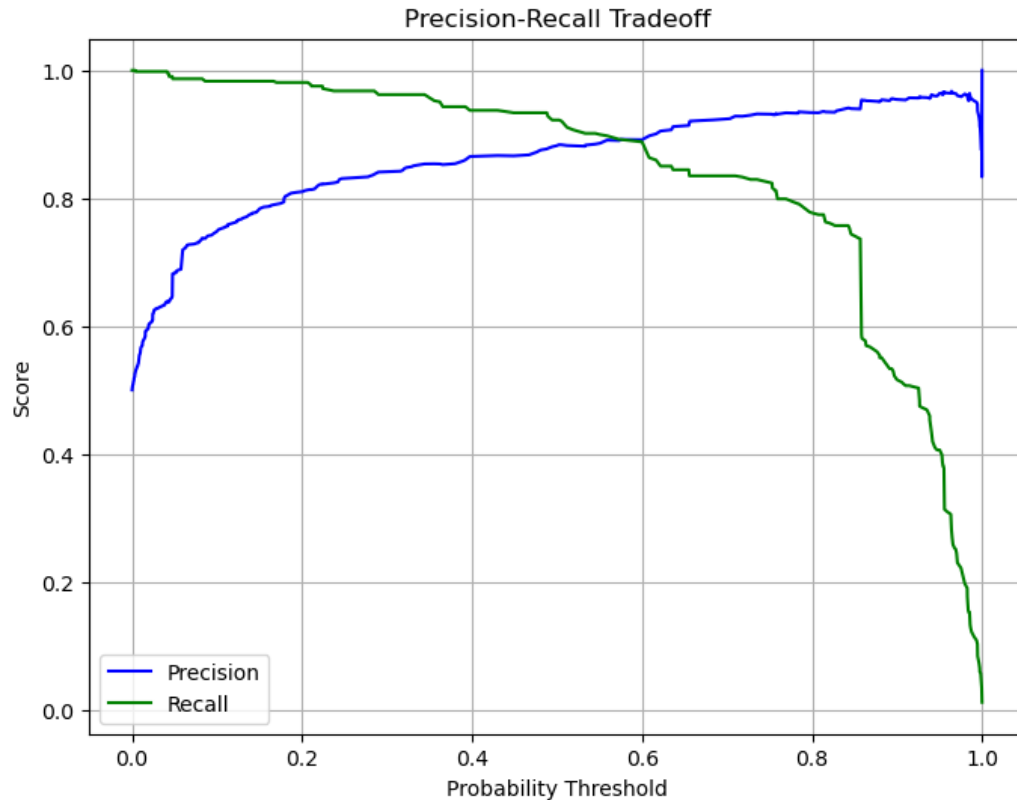
Feature Importance Graph:



Hypertuned random forest tree:



Key Insights



- **Mid-age Customers Dominate:** The majority of policyholders are between 30 and 30–45 years, and long-tenure customers are typically older, reflecting customer loyalty patterns.
- **Strong Fraud Indicators Identified:** Features like `claim_per_vehicle`, `claim_income_ratio`, `incident_hour_of_the_day`, and `insured_occupation` emerged as strong fraud predictors.
- **Multicollinearity Detected:** High correlation among `vehicle_claim`, `injury_claim`, and `property_claim` required careful feature selection to avoid model distortion.
- **Class Imbalance Addressed:** `RandomOverSampler` helped balance the fraud and non-fraud classes, improving model fairness and learning from minority class patterns.
- **Logistic Regression:** Delivered 90%+ training accuracy, with solid validation performance ($F1 = 0.67$), though slightly weaker recall than Random Forest.
- **Random Forest Outperformed in Recall:** With a validation **recall of 81%** and F1 score of **0.70**, it better captured actual fraud cases, making it more suitable for minimizing missed frauds.
- **Precision-Recall Trade-Off:** Boosting recall improves fraud detection but may slightly reduce precision—more claims are flagged as fraud, increasing false positives. Businesses must weigh tolerance for false alarms vs. missed fraud.
- **Business Takeaway:** The fraud detection system, especially using random forest, can reliably assist claim reviewers by flagging high-risk claims early, reducing manual load and financial loss.



Key Questions

- **Q.1 How can we analyze historical claim data to detect patterns that indicate fraudulent claims?**
 - By transforming and enriching historical claim data through feature engineering and simplifying categorical variables, we reveal underlying patterns (e.g., higher claim/income ratio, specific hours of incidents, certain occupations) that are strong indicators of fraud. ML models then learn from these patterns to identify future suspicious claims.
 - Perform **EDA & feature engineering** to uncover hidden trends (e.g., high claim/income ratios, late-night incidents).
 - Use ML algorithms to **learn fraud patterns** from labeled historical data.
 - **Correlate customer, vehicle, and incident variables** to discover fraud-prone profiles.
- **Q.2 Which features are the most predictive of fraudulent behavior?**
 - **Claim-based ratios:** claim_per_vehicle, claim_income_ratio
 - **Time & occupation:** incident_hour_of_the_day, insured_occupation
 - **Policy & vehicle info:** policy_csl, auto_make, auto_model
 - These features are identified using **RFECV and random forest feature importance**.

Key Questions



- **Q.3 Based on past data, can we predict the likelihood of fraud for an incoming claim?**

- Yes — predictive models trained on historical data achieved **83%+ accuracy** on unseen claims.
- **Random Forest** showed better performance in detecting fraud (higher recall and F1-score).
- Models learn complex relationships (e.g., claim patterns, behavior traits) beyond manual detection.
- Can be deployed to **flag high-risk claims in real-time**, helping underwriters prioritize effectively.
- Helps reduce **false negatives** (missed frauds) and ensures **early intervention** in the claim cycle.

- **Q.4 What insights can be drawn from the model that can help in improving the fraud detection process?**

- Claims with **high claim/income ratios or high vehicle involvement** are more suspicious.
- **Late-night incidents** had higher fraud probabilities — timing matters.
- Certain **occupations and relationships** showed recurring fraud patterns.
- Specific **vehicle makes/models** were linked to suspicious claims (e.g., frequent fraud cases in certain brands).
- Simplified categories and feature engineering improved pattern visibility.
- Focused monitoring and **dynamic risk scoring** can proactively reduce fraud payouts.
- **Feature importance transparency** allows business teams to build actionable rules and triggers.

Recommendations



Deploy the Random Forest model in the claim review pipeline to assist with early fraud detection.



Monitor high-risk claims based on model scores, especially those with late incident times or inflated claim ratios.



Regularly update models with new claim data to improve accuracy and adapt to evolving fraud patterns.



Integrate business rules using top features (e.g., occupation, vehicle make) for pre-screening before manual inspection.



Flag anomalies proactively by setting thresholds for key indicators like vehicle involvement and claim size.



Build an alert system for underwriters based on model outputs to support efficient case prioritization.



Educate internal teams on high-risk features and fraud profiles identified through the model.

Conclusion

- The fraud detection analysis for **Global Insure** successfully demonstrated the power of data-driven approaches in identifying potentially fraudulent insurance claims. Using a combination of **logistic regression** and **random forest** models, we were able to:
 - Achieve over **83% accuracy** on validation data.
 - Highlight key predictors like **claim/income ratios**, **incident timing**, and **occupation patterns**.
 - Show that **Random Forest outperformed Logistic Regression** in identifying fraud with higher recall and F1-score.
 - Develop a scalable, interpretable, and actionable model that can support real-time fraud detection.

THANK YOU

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